

Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of this section is to define a suitable Grothendieck topology on X such that we can define a structure sheaf of analytic functions on X , and then study its basic properties.

1 Affinoid domains

On a general open subset $U \subseteq X$, the ring of analytic functions on U may behave badly.

Yang: To be continued.

Definition 1. A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$ -affinoid algebra A_V and a morphism of $\mathbf{k}$ -affinoid algebras $\theta : A \rightarrow A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\psi : A \rightarrow B$ such that the induced map on spectra $\mathcal{M}(\psi) : \mathcal{M}(B) \rightarrow X$ has its image contained in V , there exists a unique bounded homomorphism $\varphi : A_V \rightarrow B$ such that the following right diagram commutes:

$$\begin{array}{ccc} & V & \\ \swarrow & & \nwarrow \\ X & \xleftarrow{\mathcal{M}(\psi)} & \mathcal{M}(B) \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} & A_V & \\ \theta \nearrow & & \searrow \varphi \\ A & \xrightarrow{\psi} & B \end{array}$$

In this case, we say that V is represented by the $\mathbf{k}$ -affinoid algebra A_V . If A_V is strict, then we say that V is a strict affinoid domain.

Slogan A closed subset $V \subset X$ is an affinoid domain if the functor “ $\text{Mor}(-, V)$ ” is representable by a $\mathbf{k}$ -affinoid algebra A_V .

There are several important classes of affinoid domains.

Construction 2. Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in A and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of X :

$$X(\underline{f/r}; \underline{s/g}) := \{x \in X : |f_i(x)| \leq r_i, |g_j(x)| \geq s_j, 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Such a closed subset is called a *Laurent domain* of X . If $m = 0$, then it is called a *Weierstrass domain* of X , denoted by

$$X(\underline{f/r}) := \{x \in X : |f_i(x)| \leq r_i, 1 \leq i \leq n\}.$$

More generally, let $f = (f_1, \dots, f_n), g$ be elements in A such that the ideal generated by them is the whole algebra A . Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following closed subset of X :

$$X(\underline{(f/g)/p}) := \{x \in X : |f_i(x)| \leq p_i |g(x)|, 1 \leq i \leq n\}.$$

| Such a closed subset is called a *rational domain* of X .

Proposition 3. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Then any Weierstrass domain, Laurent domain, or rational domain of X is an affinoid domain. Moreover, if A is strict, then these affinoid domains are all strict. *Yang: To be checked.*

| *Proof.* *Yang: To be added.* □

We have a sequence of inclusion:

$$\{\text{Affinoid domains}\} \supseteq \{\text{Rational domains}\} \supseteq \{\text{Laurent domains}\} \supseteq \{\text{Weierstrass domains}\}.$$

Yang: Why Laurent domain is rational domain?

Yang: Is these domain open?

Proposition 4. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural map on spectra $\mathcal{M}(\theta) : \mathcal{M}(A_V) \rightarrow X$ induced by the representing homomorphism $\theta : A \rightarrow A_V$ has its image equal to V , and induces a homeomorphism $\mathcal{M}(A_V) \cong V$.

| *Proof.* *Yang: To be added.* □

Proposition 5. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural bounded homomorphism $\varphi : A \rightarrow A_V$ is flat.

| *Proof.* *Yang: To be added.* □

Proposition 6. Let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the inverse image $f^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $B \hat{\otimes}_A A_V$. *Yang: To be revised.*

| *Proof.* *Yang: To be added.* □

Proposition 7. Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$ -affinoid algebras A_{V_1}, A_{V_2} , respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $A_{V_1} \hat{\otimes}_A A_{V_2}$. *Yang: To be checked.*

| *Proof.* *Yang: To be added.* □

Proposition 8. Let $x \in X$ be a point. Then the collection of strict affinoid domains of X containing x forms a fundamental system of neighborhoods of x . Explicitly, for any open neighborhood $U \subseteq X$ of x , there exists a strict affinoid domain $V \subseteq X$ such that $x \in V \subseteq U$. *Yang: To be checked.*

Theorem 9 (Gerritzen-Grauert theorem). Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \dots, V_m \subseteq X$ such that $V = V_1 \cup \dots \cup V_m$. *Yang: To be checked.*

| *Proof.* *Yang: To be added.* □

Definition 10. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Suppose that $A_V \hat{\otimes} \mathbf{K}$ is strictly $\mathbf{K}$ -affinoid for some completed valued field extension $\mathbf{K}/\mathbf{k}$. The *dimension* of

V is defined as the Krull dimension of the affinoid algebra $A_V \hat{\otimes} \mathbf{K}$:

$$\dim(V) := \dim_{\text{Krull}}(A_V \hat{\otimes} \mathbf{K}).$$

By Proposition 23, the definition is independent of the choice of $\mathbf{K}$.

2 The Grothendieck topology and structure sheaf

Theorem 11 (Acyclicity theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i \in I}$ be a finite covering of X by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j \in J} V_j$, which is an affinoid domain by Proposition 7. Then the augmented Čech complex

$$0 \rightarrow A \rightarrow \prod_{i \in I} A_{V_i} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=2}} A_{V_J} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=3}} A_{V_J} \rightarrow \cdots$$

is exact. Yang: To be checked.

Definition 12. A *special domain* of X is a finite union of affinoid domains of X . Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each V_i an affinoid domain. We define

$$A_V := \ker \left(\prod_{i=1}^m A_{V_i} \rightarrow \prod_{1 \leq i < j \leq m} A_{V_i \cap V_j} \right).$$

Yang: To be checked.

Proposition 13. Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each V_i an affinoid domain. Then V is an affinoid domain if and only if the $\mathbf{k}$ -algebra A_V defined in Definition 12 is a $\mathbf{k}$ -affinoid algebra. In this case, V is represented by the $\mathbf{k}$ -affinoid algebra A_V . Yang: To be added.

Definition 14. The *site of special domains* on X , is defined as follows:

- the underlying category has as objects the special domains $V \subseteq X$, and the morphisms are given by the inclusion maps;
- a covering of an objects $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i \in I}$ such that $V = \bigcup_{i \in I} V_i$.

The *structure sheaf on site of special domains* is given by assigning to each special domain $V \subseteq X$ the ring A_V . Yang: To be revised.

Definition 15. For every open subset $U \subseteq X$, we define the ring of analytic functions on U as follows:

$$\mathcal{O}_X(U) := \varprojlim_{V \subseteq U} A_V \in \mathbf{Alg}_{\mathbf{k}},$$

where the limit is taken over all special domains $V \subseteq U$, and A_V is the $\mathbf{k}$ -algebra representing the special domain V . Yang: To be revised.

Remark 16. Note that in [Definition 14](#) we have defined a site on X consisting of special domains, and in [Definition 15](#) we have defined a structure sheaf on the usual topological space X . In fact, these two definitions are closely related. Yang: To be revised.

Definition 17. The *category of $\mathbf{k}$ -affinoid spaces* is defined as the opposite category of the category of $\mathbf{k}$ -affinoid algebras:

$$\mathbf{k} := (\mathbf{Alg}_{\mathbf{k}}^{\text{aff}})^{\text{op}}.$$

Yang: To be revised.

Yang: Coherent sheaf.

3 Boundary and interior

Let A, B be two $\mathbf{k}$ -affinoid algebras, and let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$.

Definition 18. The *relative interior* of Y over X is defined as

$$(Y/X) := \{y \in Y : \text{there exists an admissible surjective } A \rightarrow B \text{ inducing } y\}.$$

Yang: To be revised.

Definition 19.

4 Some local properties

Yang: The local ring $\mathcal{O}_{X,x}$.

Yang: The residue field and its completion.

Example 20. Yang: To be added.

Proposition 21. The local ring $\mathcal{O}_{X,x}$ at any point $x \in X$ is noetherian.

Definition 22. We say that the affinoid space X has a local property $\mathcal{P}$ at a point $x \in X$ if the local ring $\mathcal{O}_{X,x}$ has the property $\mathcal{P}$. Yang: To be added.

Appendix

Proposition 23. Let A be a strictly $\mathbf{k}$ -affinoid algebra. For any completed valued field extension $\mathbf{K}/\mathbf{k}$, we have

$$\dim_{\mathbf{K}\text{rull}}(A) = \dim_{\mathbf{K}\text{rull}}(A \hat{\otimes}_{\mathbf{k}} \mathbf{K}).$$

Yang: To be checked.